# MSG No. 223.109 $P_Im\bar{3}n$ [ Type IV, cubic ]

Table 1: Wyckoff site: 2a, site symmetry: m-3m'

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,8,9,10,17,18,19,20,21,22,23,24,25,32,33,34,41,42,43,44,45,46,47,48, |
| | | 50,51,52,53,54,55,59,60,61,62,63,64,74,75,76,77,78,79,83,84,85,86,87,88] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [2,3,4,5,6,7,11,12,13,14,15,16,26,27,28,29,30,31,35,36,37,38,39,40, |
| | | 49,56,57,58,65,66,67,68,69,70,71,72,73,80,81,82,89,90,91,92,93,94,95,96] |

Table 2: Wyckoff site: 6b, site symmetry: 4'/mm.m'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [1,8,9,10,25,32,33,34,50,51,61,62,74,75,85,86] |
| 2 | $\left[\frac{1}{2}, 0, 0\right]$ | [2,3,13,14,26,27,37,38,49,56,57,58,73,80,81,82] |
| 3 | $\left[0, 0, \frac{1}{2}\right]$ | [4,5,15,16,28,29,39,40,66,67,69,71,90,91,93,95] |
| 4 | $\left[0, \frac{1}{2}, 0\right]$ | [6,7,11,12,30,31,35,36,65,68,70,72,89,92,94,96] |
| 5 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [17,20,22,24,41,44,46,48,54,55,59,60,78,79,83,84] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [18,19,21,23,42,43,45,47,52,53,63,64,76,77,87,88] |

Table 3: Wyckoff site: 8c, site symmetry: .-3'm'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,12,14,16,17,18,73,84,86,88,89,90] |
| 2 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [2,7,9,15,19,24,74,79,81,87,91,96] |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [3,4,10,11,20,21,75,76,82,83,92,93] |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [5,6,8,13,22,23,77,78,80,85,94,95] |
| 5 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [25,36,38,40,41,42,49,60,62,64,65,66] |
| 6 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [26,31,33,39,43,48,50,55,57,63,67,72] |
| 7 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [27,28,34,35,44,45,51,52,58,59,68,69] |
| 8 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [29,30,32,37,46,47,53,54,56,61,70,71] |

Table 4: Wyckoff site: 12d, site symmetry: -4m.2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, 0, \frac{1}{2}\right]$ | [1,8,13,14,26,27,33,34] |
| 2 | $\left[\frac{3}{4}, 0, \frac{1}{2}\right]$ | [2,3,9,10,25,32,37,38] |
| 3 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [4,16,18,21,29,39,43,47] |
| 4 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [5,15,19,23,28,40,42,45] |
| 5 | $\left[\frac{1}{2}, \frac{3}{4}, 0\right]$ | [6,11,20,22,31,36,41,48] |
| 6 | $\left[\frac{1}{2}, \frac{1}{4}, 0\right]$ | [7,12,17,24,30,35,44,46] |
| 7 | $\left[\frac{3}{4}, \frac{1}{2}, 0\right]$ | [49,56,61,62,74,75,81,82] |
| 8 | $\left[\frac{1}{4}, \frac{1}{2}, 0\right]$ | [50,51,57,58,73,80,85,86] |
| 9 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [52,64,66,69,77,87,91,95] |
| 10 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [53,63,67,71,76,88,90,93] |
| 11 | $\left[0, \frac{1}{4}, \frac{1}{2}\right]$ | [54,59,68,70,79,84,89,96] |

*continued ...*

Table 4

| No. | position | mapping |
|---|---|---|
| 12 | $\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[55,60,65,72,78,83,92,94]` |

Table 5: Wyckoff site: `12e`, site symmetry: `4'm.m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,8,33,34,50,51,85,86]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,37,38,49,56,81,82]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$ | `[4,16,29,39,67,71,90,93]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$ | `[5,15,28,40,66,69,91,95]` |
| 5 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[6,11,31,36,65,72,92,94]` |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[7,12,30,35,68,70,89,96]` |
| 7 | $[-x,\ 0,\ 0]$ | `[9,10,25,32,61,62,74,75]` |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[13,14,26,27,57,58,73,80]` |
| 9 | $[0,\ x,\ 0]$ | `[17,24,44,46,54,59,79,84]` |
| 10 | $[0,\ 0,\ x]$ | `[18,21,43,47,53,63,76,88]` |
| 11 | $[0,\ 0,\ -x]$ | `[19,23,42,45,52,64,77,87]` |
| 12 | $[0,\ -x,\ 0]$ | `[20,22,41,48,55,60,78,83]` |

Table 6: Wyckoff site: `16f`, site symmetry: `.3m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ x]$ | `[1,17,18,84,86,88]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | `[2,7,15,81,91,96]` |
| 3 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[3,4,11,82,92,93]` |
| 4 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[5,6,13,80,94,95]` |
| 5 | $[x,\ -x,\ -x]$ | `[8,22,23,77,78,85]` |
| 6 | $[-x,\ x,\ -x]$ | `[9,19,24,74,79,87]` |
| 7 | $[-x,\ -x,\ x]$ | `[10,20,21,75,76,83]` |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | `[12,14,16,73,89,90]` |
| 9 | $[-x,\ -x,\ -x]$ | `[25,41,42,60,62,64]` |
| 10 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[26,31,39,57,67,72]` |
| 11 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | `[27,28,35,58,68,69]` |
| 12 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | `[29,30,37,56,70,71]` |
| 13 | $[-x,\ x,\ x]$ | `[32,46,47,53,54,61]` |
| 14 | $[x,\ -x,\ x]$ | `[33,43,48,50,55,63]` |
| 15 | $[x,\ x,\ -x]$ | `[34,44,45,51,52,59]` |
| 16 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[36,38,40,49,65,66]` |

Table 7: Wyckoff site: `24g`, site symmetry: `2mm..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[1,8,33,34]` |
| 2 | $\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[2,3,37,38]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 3 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$ | [4,16,29,39] |
| 4 | $\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$ | [5,15,28,40] |
| 5 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$ | [6,11,31,36] |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | [7,12,30,35] |
| 7 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | [9,10,25,32] |
| 8 | $\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$ | [13,14,26,27] |
| 9 | $\left[\frac{1}{2},\ x,\ 0\right]$ | [17,24,44,46] |
| 10 | $\left[0,\ \frac{1}{2},\ x\right]$ | [18,21,43,47] |
| 11 | $\left[0,\ \frac{1}{2},\ -x\right]$ | [19,23,42,45] |
| 12 | $\left[\frac{1}{2},\ -x,\ 0\right]$ | [20,22,41,48] |
| 13 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | [49,56,81,82] |
| 14 | $\left[x,\ \frac{1}{2},\ 0\right]$ | [50,51,85,86] |
| 15 | $\left[\frac{1}{2},\ 0,\ -x\right]$ | [52,64,77,87] |
| 16 | $\left[\frac{1}{2},\ 0,\ x\right]$ | [53,63,76,88] |
| 17 | $\left[0,\ x,\ \frac{1}{2}\right]$ | [54,59,79,84] |
| 18 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | [55,60,78,83] |
| 19 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | [57,58,73,80] |
| 20 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | [61,62,74,75] |
| 21 | $\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [65,72,92,94] |
| 22 | $\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$ | [66,69,91,95] |
| 23 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$ | [67,71,90,93] |
| 24 | $\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [68,70,89,96] |

Table 8: Wyckoff site: 24h, site symmetry: m.2'm'

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ y]$ | [1,32,61,86] |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$ | [2,27,58,81] |
| 3 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [3,26,57,82] |
| 4 | $\left[y+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [4,40,66,93] |
| 5 | $\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [5,39,67,95] |
| 6 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$ | [6,35,68,94] |
| 7 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ y+\frac{1}{2}\right]$ | [7,36,65,96] |
| 8 | $[0,\ -y,\ -y]$ | [8,25,62,85] |
| 9 | $[0,\ y,\ -y]$ | [9,34,51,74] |
| 10 | $[0,\ -y,\ y]$ | [10,33,50,75] |
| 11 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [11,30,70,92] |
| 12 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [12,31,72,89] |
| 13 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [13,38,49,80] |
| 14 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$ | [14,37,56,73] |
| 15 | $\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [15,29,71,91] |
| 16 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [16,28,69,90] |
| 17 | $[y,\ 0,\ y]$ | [17,48,55,84] |
| 18 | $[y,\ y,\ 0]$ | [18,45,52,88] |
| 19 | $[-y,\ y,\ 0]$ | [19,47,53,87] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 20 | $[-y,\ 0,\ y]$ | `[20,46,54,83]` |
| 21 | $[-y,\ -y,\ 0]$ | `[21,42,64,76]` |
| 22 | $[y,\ 0,\ -y]$ | `[22,44,59,78]` |
| 23 | $[y,\ -y,\ 0]$ | `[23,43,63,77]` |
| 24 | $[-y,\ 0,\ -y]$ | `[24,41,60,79]` |

Table 9: Wyckoff site: `48i`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$ | `[1,14]` |
| 2 | $\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$ | `[2,9]` |
| 3 | $\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$ | `[3,10]` |
| 4 | $\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | `[4,21]` |
| 5 | $\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | `[5,23]` |
| 6 | $\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$ | `[6,22]` |
| 7 | $\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$ | `[7,24]` |
| 8 | $\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$ | `[8,13]` |
| 9 | $\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$ | `[11,20]` |
| 10 | $\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$ | `[12,17]` |
| 11 | $\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | `[15,19]` |
| 12 | $\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | `[16,18]` |
| 13 | $\left[\frac{3}{4},\ -y,\ y+\frac{1}{2}\right]$ | `[25,38]` |
| 14 | $\left[\frac{1}{4},\ -y,\ \frac{1}{2}-y\right]$ | `[26,33]` |
| 15 | $\left[\frac{1}{4},\ y,\ y+\frac{1}{2}\right]$ | `[27,34]` |
| 16 | $\left[y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | `[28,45]` |
| 17 | $\left[-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | `[29,47]` |
| 18 | $\left[y+\frac{1}{2},\ \frac{1}{4},\ y\right]$ | `[30,46]` |
| 19 | $\left[\frac{1}{2}-y,\ \frac{3}{4},\ y\right]$ | `[31,48]` |
| 20 | $\left[\frac{3}{4},\ y,\ \frac{1}{2}-y\right]$ | `[32,37]` |
| 21 | $\left[\frac{1}{2}-y,\ \frac{1}{4},\ -y\right]$ | `[35,44]` |
| 22 | $\left[y+\frac{1}{2},\ \frac{3}{4},\ -y\right]$ | `[36,41]` |
| 23 | $\left[y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | `[39,43]` |
| 24 | $\left[-y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | `[40,42]` |
| 25 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$ | `[49,62]` |
| 26 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ y\right]$ | `[50,57]` |
| 27 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ -y\right]$ | `[51,58]` |
| 28 | $\left[\frac{1}{2}-y,\ y,\ \frac{3}{4}\right]$ | `[52,69]` |
| 29 | $\left[y+\frac{1}{2},\ y,\ \frac{1}{4}\right]$ | `[53,71]` |
| 30 | $\left[-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$ | `[54,70]` |
| 31 | $\left[y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$ | `[55,72]` |
| 32 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ y\right]$ | `[56,61]` |
| 33 | $\left[y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$ | `[59,68]` |
| 34 | $\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$ | `[60,65]` |
| 35 | $\left[\frac{1}{2}-y,\ -y,\ \frac{1}{4}\right]$ | `[63,67]` |
| 36 | $\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$ | `[64,66]` |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 37 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ y\right]$ | [73,86] |
| 38 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ -y\right]$ | [74,81] |
| 39 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ y\right]$ | [75,82] |
| 40 | $\left[y+\frac{1}{2},\ -y,\ \frac{1}{4}\right]$ | [76,93] |
| 41 | $\left[\frac{1}{2}-y,\ -y,\ \frac{3}{4}\right]$ | [77,95] |
| 42 | $\left[y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$ | [78,94] |
| 43 | $\left[-y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$ | [79,96] |
| 44 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ -y\right]$ | [80,85] |
| 45 | $\left[-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$ | [83,92] |
| 46 | $\left[y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$ | [84,89] |
| 47 | $\left[y+\frac{1}{2},\ y,\ \frac{3}{4}\right]$ | [87,91] |
| 48 | $\left[\frac{1}{2}-y,\ y,\ \frac{1}{4}\right]$ | [88,90] |

Table 10: Wyckoff site: 48j, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | [1,32] |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | [2,27] |
| 3 | $\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [3,26] |
| 4 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [4,40] |
| 5 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [5,39] |
| 6 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [6,35] |
| 7 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [7,36] |
| 8 | $[0,\ -y,\ -z]$ | [8,25] |
| 9 | $[0,\ y,\ -z]$ | [9,34] |
| 10 | $[0,\ -y,\ z]$ | [10,33] |
| 11 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [11,30] |
| 12 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [12,31] |
| 13 | $\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [13,38] |
| 14 | $\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | [14,37] |
| 15 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [15,29] |
| 16 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [16,28] |
| 17 | $[z,\ 0,\ y]$ | [17,48] |
| 18 | $[y,\ z,\ 0]$ | [18,45] |
| 19 | $[-y,\ z,\ 0]$ | [19,47] |
| 20 | $[-z,\ 0,\ y]$ | [20,46] |
| 21 | $[-y,\ -z,\ 0]$ | [21,42] |
| 22 | $[z,\ 0,\ -y]$ | [22,44] |
| 23 | $[y,\ -z,\ 0]$ | [23,43] |
| 24 | $[-z,\ 0,\ -y]$ | [24,41] |
| 25 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [49,80] |
| 26 | $[0,\ -z,\ y]$ | [50,75] |
| 27 | $[0,\ z,\ -y]$ | [51,74] |
| 28 | $[z,\ y,\ 0]$ | [52,88] |
| 29 | $[-z,\ y,\ 0]$ | [53,87] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 30 | $[-y,\ 0,\ z]$ | [54,83] |
| 31 | $[y,\ 0,\ z]$ | [55,84] |
| 32 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [56,73] |
| 33 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [57,82] |
| 34 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [58,81] |
| 35 | $[y,\ 0,\ -z]$ | [59,78] |
| 36 | $[-y,\ 0,\ -z]$ | [60,79] |
| 37 | $[0,\ z,\ y]$ | [61,86] |
| 38 | $[0,\ -z,\ -y]$ | [62,85] |
| 39 | $[z,\ -y,\ 0]$ | [63,77] |
| 40 | $[-z,\ -y,\ 0]$ | [64,76] |
| 41 | $\left[z+\frac{1}{2},\ \frac{1}{2},\ y+\frac{1}{2}\right]$ | [65,96] |
| 42 | $\left[y+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}\right]$ | [66,93] |
| 43 | $\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ \frac{1}{2}\right]$ | [67,95] |
| 44 | $\left[\frac{1}{2}-z,\ \frac{1}{2},\ y+\frac{1}{2}\right]$ | [68,94] |
| 45 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{1}{2}\right]$ | [69,90] |
| 46 | $\left[z+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [70,92] |
| 47 | $\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}\right]$ | [71,91] |
| 48 | $\left[\frac{1}{2}-z,\ \frac{1}{2},\ \frac{1}{2}-y\right]$ | [72,89] |

Table 11: Wyckoff site: 48k, site symmetry: ..m'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | [1,84] |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$ | [2,91] |
| 3 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [3,93] |
| 4 | $\left[z+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [4,92] |
| 5 | $\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [5,94] |
| 6 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [6,80] |
| 7 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [7,81] |
| 8 | $[x,\ -x,\ -z]$ | [8,78] |
| 9 | $[-x,\ x,\ -z]$ | [9,79] |
| 10 | $[-x,\ -x,\ z]$ | [10,83] |
| 11 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [11,82] |
| 12 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [12,73] |
| 13 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [13,95] |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$ | [14,90] |
| 15 | $\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | [15,96] |
| 16 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | [16,89] |
| 17 | $[z,\ x,\ x]$ | [17,88] |
| 18 | $[x,\ z,\ x]$ | [18,86] |
| 19 | $[-x,\ z,\ -x]$ | [19,74] |
| 20 | $[-z,\ -x,\ x]$ | [20,76] |
| 21 | $[-x,\ -z,\ x]$ | [21,75] |
| 22 | $[z,\ -x,\ -x]$ | [22,77] |

*continued ...*

Table 11

| No. | position | mapping |
|---|---|---|
| 23 | $[x,\,-z,\,-x]$ | [23,85] |
| 24 | $[-z,\,x,\,-x]$ | [24,87] |
| 25 | $[-x,\,-x,\,-z]$ | [25,60] |
| 26 | $\left[\frac{1}{2}-x,\,z+\frac{1}{2},\,\frac{1}{2}-x\right]$ | [26,67] |
| 27 | $\left[\frac{1}{2}-x,\,\frac{1}{2}-z,\,x+\frac{1}{2}\right]$ | [27,69] |
| 28 | $\left[\frac{1}{2}-z,\,\frac{1}{2}-x,\,x+\frac{1}{2}\right]$ | [28,68] |
| 29 | $\left[z+\frac{1}{2},\,\frac{1}{2}-x,\,\frac{1}{2}-x\right]$ | [29,70] |
| 30 | $\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$ | [30,56] |
| 31 | $\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,\frac{1}{2}-z\right]$ | [31,57] |
| 32 | $[-x,\,x,\,z]$ | [32,54] |
| 33 | $[x,\,-x,\,z]$ | [33,55] |
| 34 | $[x,\,x,\,-z]$ | [34,59] |
| 35 | $\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$ | [35,58] |
| 36 | $\left[x+\frac{1}{2},\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$ | [36,49] |
| 37 | $\left[x+\frac{1}{2},\,\frac{1}{2}-z,\,\frac{1}{2}-x\right]$ | [37,71] |
| 38 | $\left[x+\frac{1}{2},\,z+\frac{1}{2},\,x+\frac{1}{2}\right]$ | [38,66] |
| 39 | $\left[\frac{1}{2}-z,\,x+\frac{1}{2},\,\frac{1}{2}-x\right]$ | [39,72] |
| 40 | $\left[z+\frac{1}{2},\,x+\frac{1}{2},\,x+\frac{1}{2}\right]$ | [40,65] |
| 41 | $[-z,\,-x,\,-x]$ | [41,64] |
| 42 | $[-x,\,-z,\,-x]$ | [42,62] |
| 43 | $[x,\,-z,\,x]$ | [43,50] |
| 44 | $[z,\,x,\,-x]$ | [44,52] |
| 45 | $[x,\,z,\,-x]$ | [45,51] |
| 46 | $[-z,\,x,\,x]$ | [46,53] |
| 47 | $[-x,\,z,\,x]$ | [47,61] |
| 48 | $[z,\,-x,\,x]$ | [48,63] |

Table 12: Wyckoff site: 96l, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\,y,\,z]$ | [1] |
| 2 | $\left[x+\frac{1}{2},\,\frac{1}{2}-z,\,y+\frac{1}{2}\right]$ | [2] |
| 3 | $\left[x+\frac{1}{2},\,z+\frac{1}{2},\,\frac{1}{2}-y\right]$ | [3] |
| 4 | $\left[z+\frac{1}{2},\,y+\frac{1}{2},\,\frac{1}{2}-x\right]$ | [4] |
| 5 | $\left[\frac{1}{2}-z,\,y+\frac{1}{2},\,x+\frac{1}{2}\right]$ | [5] |
| 6 | $\left[\frac{1}{2}-y,\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$ | [6] |
| 7 | $\left[y+\frac{1}{2},\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$ | [7] |
| 8 | $[x,\,-y,\,-z]$ | [8] |
| 9 | $[-x,\,y,\,-z]$ | [9] |
| 10 | $[-x,\,-y,\,z]$ | [10] |
| 11 | $\left[y+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}-z\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$ | [12] |
| 13 | $\left[\frac{1}{2}-x,\,z+\frac{1}{2},\,y+\frac{1}{2}\right]$ | [13] |
| 14 | $\left[\frac{1}{2}-x,\,\frac{1}{2}-z,\,\frac{1}{2}-y\right]$ | [14] |
| 15 | $\left[z+\frac{1}{2},\,\frac{1}{2}-y,\,x+\frac{1}{2}\right]$ | [15] |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 16 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | [16] |
| 17 | $[z,\ x,\ y]$ | [17] |
| 18 | $[y,\ z,\ x]$ | [18] |
| 19 | $[-y,\ z,\ -x]$ | [19] |
| 20 | $[-z,\ -x,\ y]$ | [20] |
| 21 | $[-y,\ -z,\ x]$ | [21] |
| 22 | $[z,\ -x,\ -y]$ | [22] |
| 23 | $[y,\ -z,\ -x]$ | [23] |
| 24 | $[-z,\ x,\ -y]$ | [24] |
| 25 | $[-x,\ -y,\ -z]$ | [25] |
| 26 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [26] |
| 27 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | [27] |
| 28 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | [28] |
| 29 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | [29] |
| 30 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [30] |
| 31 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [31] |
| 32 | $[-x,\ y,\ z]$ | [32] |
| 33 | $[x,\ -y,\ z]$ | [33] |
| 34 | $[x,\ y,\ -z]$ | [34] |
| 35 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [35] |
| 36 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [36] |
| 37 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | [37] |
| 38 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | [38] |
| 39 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [39] |
| 40 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [40] |
| 41 | $[-z,\ -x,\ -y]$ | [41] |
| 42 | $[-y,\ -z,\ -x]$ | [42] |
| 43 | $[y,\ -z,\ x]$ | [43] |
| 44 | $[z,\ x,\ -y]$ | [44] |
| 45 | $[y,\ z,\ -x]$ | [45] |
| 46 | $[-z,\ x,\ y]$ | [46] |
| 47 | $[-y,\ z,\ x]$ | [47] |
| 48 | $[z,\ -x,\ y]$ | [48] |
| 49 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [49] |
| 50 | $[x,\ -z,\ y]$ | [50] |
| 51 | $[x,\ z,\ -y]$ | [51] |
| 52 | $[z,\ y,\ -x]$ | [52] |
| 53 | $[-z,\ y,\ x]$ | [53] |
| 54 | $[-y,\ x,\ z]$ | [54] |
| 55 | $[y,\ -x,\ z]$ | [55] |
| 56 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [56] |
| 57 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [57] |
| 58 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [58] |
| 59 | $[y,\ x,\ -z]$ | [59] |
| 60 | $[-y,\ -x,\ -z]$ | [60] |
| 61 | $[-x,\ z,\ y]$ | [61] |
| 62 | $[-x,\ -z,\ -y]$ | [62] |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 63 | $[z, -y, x]$ | [63] |
| 64 | $[-z, -y, -x]$ | [64] |
| 65 | $\left[z+\frac{1}{2}, x+\frac{1}{2}, y+\frac{1}{2}\right]$ | [65] |
| 66 | $\left[y+\frac{1}{2}, z+\frac{1}{2}, x+\frac{1}{2}\right]$ | [66] |
| 67 | $\left[\frac{1}{2}-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$ | [67] |
| 68 | $\left[\frac{1}{2}-z, \frac{1}{2}-x, y+\frac{1}{2}\right]$ | [68] |
| 69 | $\left[\frac{1}{2}-y, \frac{1}{2}-z, x+\frac{1}{2}\right]$ | [69] |
| 70 | $\left[z+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-y\right]$ | [70] |
| 71 | $\left[y+\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-x\right]$ | [71] |
| 72 | $\left[\frac{1}{2}-z, x+\frac{1}{2}, \frac{1}{2}-y\right]$ | [72] |
| 73 | $\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}-z\right]$ | [73] |
| 74 | $[-x, z, -y]$ | [74] |
| 75 | $[-x, -z, y]$ | [75] |
| 76 | $[-z, -y, x]$ | [76] |
| 77 | $[z, -y, -x]$ | [77] |
| 78 | $[y, -x, -z]$ | [78] |
| 79 | $[-y, x, -z]$ | [79] |
| 80 | $\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [80] |
| 81 | $\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [81] |
| 82 | $\left[x+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-z\right]$ | [82] |
| 83 | $[-y, -x, z]$ | [83] |
| 84 | $[y, x, z]$ | [84] |
| 85 | $[x, -z, -y]$ | [85] |
| 86 | $[x, z, y]$ | [86] |
| 87 | $[-z, y, -x]$ | [87] |
| 88 | $[z, y, x]$ | [88] |
| 89 | $\left[\frac{1}{2}-z, \frac{1}{2}-x, \frac{1}{2}-y\right]$ | [89] |
| 90 | $\left[\frac{1}{2}-y, \frac{1}{2}-z, \frac{1}{2}-x\right]$ | [90] |
| 91 | $\left[y+\frac{1}{2}, \frac{1}{2}-z, x+\frac{1}{2}\right]$ | [91] |
| 92 | $\left[z+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-y\right]$ | [92] |
| 93 | $\left[y+\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-x\right]$ | [93] |
| 94 | $\left[\frac{1}{2}-z, x+\frac{1}{2}, y+\frac{1}{2}\right]$ | [94] |
| 95 | $\left[\frac{1}{2}-y, z+\frac{1}{2}, x+\frac{1}{2}\right]$ | [95] |
| 96 | $\left[z+\frac{1}{2}, \frac{1}{2}-x, y+\frac{1}{2}\right]$ | [96] |